

Jalpan Vyas

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EDUCATION

- Karnavati University (Unitedworld Institute of Technology)** Ahmedabad, Gujarat, India
Bachelor of Technology - Computer Science; CGPA: 7.2
Aug 2023 - July 2027
Courses: Operating Systems, Data Structures, Analysis of Algorithms, Artificial Intelligence, Machine Learning, Networking, Databases

SKILLS SUMMARY

- Languages:** Python, JavaScript, TypeScript, Rust, C#, C, SQL, Shell, HTML5, CSS3, R
- Frameworks:** TensorFlow, PyTorch, Scikit-learn, LangChain, LlamaIndex, React, Next.js, FastAPI, Django, OpenCV
- Tools:** Git, Docker, PostgreSQL, MongoDB, Qdrant, Ollama, n8n, FFmpeg, Jupyter
- Platforms:** Linux, AWS, GitHub, Hugging Face
- Soft Skills:** Problem Solving, Analytical Thinking, Adaptability, Technical Writing, Time Management

EXPERIENCE

- Bhaskaracharya Institute for Space Applications and Geoinformatics** Gandhinagar, Gujarat
AI/ML Intern (On-site)
May 2026 - Aug 2026
 - Enterprise RAG Architecture:** Architected DocRAFT, a local-first Retrieval-Augmented Generation system using Qdrant, Ollama, FastAPI, and Next.js for intelligent semantic search.
 - Multimodal Document Ingestion:** Executed layout-aware PDF parsing and asynchronous processing via FastAPI background tasks, ensuring non-blocking execution for heavy technical documents.
 - Vision Intelligence & OCR:** Engineered a two-pass image analysis pipeline using RapidOCR and Granite Vision models to extract labels and generate AI descriptions, making visual data semantically searchable.
 - Semantic Chunking Strategy:** Optimized retrieval fidelity by utilizing LlamaIndex for heading-aware node parsing, preventing mid-row table splits and preserving structural context up to 1500 characters.

PROJECTS

- DocRAFT (Python, FastAPI, Next.js, Qdrant, Ollama, LlamaIndex):** Developed an enterprise-grade, local-first multimodal RAG system. Formulated a two-pass image analysis pipeline using RapidOCR and Vision AI, combined with layout-aware PDF parsing and semantic chunking for precise retrieval.
- Gujarati Author Attribution (Python, NLP, Machine Learning):** Built an end-to-end NLP pipeline for Authorship Attribution on Gujarati literature using Python. Leveraged stylometry and classification models to identify stylistic signatures of regional authors.
- Aspis Engine (Python, PySide6, Pygame, Pymunk):** Designed a cross-platform 2D game engine featuring a custom visual editor built with PySide6. Applied an Entity-Component-System (ECS) architecture for highly decoupled game logic, robust rigid body physics, and real-time script hot-reloading.
- Smash (Rust, Python, NLP, Shell Development):** Created a cross-platform shell implemented in Rust with an embedded AI assistant. Integrated Natural Language Processing to seamlessly translate user requests into executable shell commands.
- Mainbranch (TypeScript, React, Next.js, Node.js):** Programmed a comprehensive Social Platform for developers featuring portfolio generation and community engagement tools. Secured RESTful APIs for user authentication and structured data persistence for scalability.

PUBLICATIONS

- LEKH-Net: A Dual-Branch Neural Architecture for Computational Stylistics:** Accepted at IEEE CIECS (July 2026). Constructed a 1D-CNN + Transformer architecture achieving 90.6% accuracy on low-resource Indic text datasets.
- ExecDraw: A Structural Program Bottleneck for Compositional Generalization:** Accepted at IEEE CICON (Oct 2026). Addressed structural bottlenecks to enhance compositional generalization in generative vision models.
- TinySLR: Bridging the Accuracy-Efficiency Gap in Static Sign Language Recognition:** Produced an edge-AI model compressed to 0.65 MB via dynamic quantization, achieving 97.80% test accuracy on sub-milliwatt hardware.

CERTIFICATIONS

- AWS Academy Graduate (Cloud Security & Web Application Builder) - Oct 2025 to March 2026
- Fundamentals of MCP - Hugging Face - February 2026
- OpenCV Bootcamp - OpenCV University - March 2026
- Intro to Deep Learning & Machine Learning - Kaggle - Nov 2025
- MongoDB Python Developer Path - MongoDB - March 2026
- Red Hat Training: Getting Started with Linux Fundamentals (RH104) - March 2026